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Inventor(s)

SAITO; Hideki et al.

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## CONNECTOR

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### Abstract

A connector includes a housing configured to form an assembling space along a connection direction, and a position assurance member to be inserted into the assembling space and assembled to the housing in such a manner as to be movable along the connection direction from a tentative lock position to a regular lock position, in which the assembling space is formed between two side walls formed in the housing and extending in the connection direction, the housing includes a lock member that engages with the position assurance member assembled to the assembling space, and locks the position assurance member at the regular lock position, the lock member includes an installation portion installed between the two side walls, and the installation portion is connected to the two side walls, and connected to a columnar portion extending from a floor portion of the assembling space, to be supported at three points.

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**Inventors:** SAITO; Hideki (Shizuoka, JP), Hasegawa; Takuya (Shizuoka, JP), Terao; Kazuya (Shizuoka, JP)

**Applicant:** Yazaki Corporation (Tokyo, JP)

**Family ID:** 96738315

**Assignee:** Yazaki Corporation (Tokyo, JP)

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-031909 filed in Japan on Mar. 4, 2024.

### BACKGROUND OF THE INVENTION

#### 1. Field of the Invention

[0002] The present invention relates to a connector.

#### 2. Description of the Related Art

[0003] As a technique related to connectors, there has been conventionally known a connector that is a connector connected to a counterpart connector, and has a connector position assurance function, as described in JP 2017-157454 A, for example. This connector includes a position assurance member that assures a fitted state with the counterpart connector. The position assurance member is attached to a housing of the connector, and arranged at a standby position. After the connector and the counterpart connector are fitted with each other, the position assurance member is moved in a fitting direction, and moved from the standby position to a fitting lock position. If the position assurance member can be moved to the fitting lock position, it is assured that the connector and the counterpart connector are appropriately fitted with each other.

[0004] The above-described connector includes a portion in which the housing and the position assurance member are engaged, and the mechanical strength of this engaged portion sometimes becomes lower as compared with other portions. Thus, the development of a connector that can achieve improvement in strength has been demanded.

### SUMMARY OF THE INVENTION

[0005] In view of the foregoing, the present invention aims to provide a connector that can achieve improvement in strength.

[0006] In order to achieve the above mentioned object, a connector according to one aspect of the present invention includes a housing configured to be connectable to a counterpart connector, and form an assembling space along a connection direction; and a position assurance member to be inserted into the assembling space and assembled to the housing in such a manner as to be movable along the connection direction from a tentative lock position to a regular lock position, wherein the assembling space is formed between two side walls formed in the housing and extending in the connection direction, the housing includes a lock member that engages with the position assurance member assembled to the assembling space, and locks the position assurance member at the regular lock position, the lock member includes an installation portion installed between the two side walls, and the installation portion is connected to the two side walls, and connected to a columnar portion extending from a floor portion of the assembling space, to be supported at three points.

[0007] The above and other objects, features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed description of presently preferred embodiments of the invention, when considered in connection with the accompanying drawings.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is an exploded perspective view of a connector according to an embodiment;

[0009] FIG. 2 is a perspective view of a connector according to an embodiment;

[0010] FIG. 3 is a plan view of a disassembled state of a connector according to an embodiment;

[0011] FIG. 4 is a perspective view of a housing in a connector according to an embodiment;

[0012] FIG. 5 is an explanatory diagram of a regularly-locked state of a position assurance member of a connector according to an embodiment;

[0013] FIG. 6 is an explanatory diagram of connection in a connector according to an embodiment; and

[0014] FIG. 7 is an explanatory diagram of mechanical strength in a connector according to an embodiment.

## DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0015] An embodiment according to the present invention will be described in detail below based on the drawings. In addition, the present invention is not limited by the embodiment. Further, components in the following embodiment include the ones that can be replaced by those skilled in the art, and are easy, or the ones that are substantially identical.

### Embodiment

[0016] The present embodiment relates to a connector. In the following description, among a first direction, a second direction, and a third direction that intersect with each other, the first direction will be referred to as a “connection direction X”, the second direction will be referred to as a “width direction Y”, and the third direction will be referred to as a “height direction Z”. Here, the connection direction X, the width direction Y, and the height direction Z are orthogonal to each other. The connection direction X corresponds to a connection direction or a fitting direction of a connector and a counterpart connector. The width direction Y corresponds to a width direction of a connector. The width direction Y and the height direction Z correspond to an orthogonal direction orthogonal to the connection direction X. Further, each direction to be used in the following direction refers to a direction in a state in which components are assembled to each other, unless otherwise specified. In addition, orthogonal here includes an approximately orthogonal state.

[0017] As illustrated in FIGS. 1 to 3, a connector 1 according to the present embodiment is a connector to be used with being connected to a counterpart connector 10, and has a connector position assurance (CPA) function. The connector 1 includes a housing 2 and a position assurance member 3. FIG. 1 illustrates a state in which the connector 1 and the counterpart connector 10 are separated, and the housing 2 and the position assurance member 3 are separated. The housing 2 is a member to be fitted with the counterpart connector 10, and is formed in a block shape, for example. The housing 2 is made of resin, for example, and is formed by resin molding. In the housing 2, an installation hole 21 for arranging a terminal is formed. The installation hole 21 is formed along the connection direction X with the counterpart connector 10. A plurality of installation holes 21 is formed, and the plurality of installation holes 21 is arrayed toward the width direction Y, for example. The installation hole 21 is formed with penetrating through the housing 2 toward the connection direction X, and a terminal of an electric wire (not illustrated) equipped with a terminal is inserted therein and held therein.

[0018] In the housing 2, an assembling space 22 is formed above a region in which the installation hole 21 is formed. The assembling space 22 is a space for inserting and assembling the position assurance member 3, and is formed between two side walls 23 provided with being separated in the width direction Y of the housing 2, for example. The side walls 23 are wall members respectively provided at ends in the width direction Y of the housing 2, and extending in the height direction Z and the connection direction X. In other words, the two side walls 23 define the assembling space 22 by partitioning side portions on both sides of the assembling space 22.

[0019] A guide portion 222 is formed in the assembling space 22. The guide portion 222 is formed

at a position at a lower end of the assembling space **22** along the connection direction X. For example, the guide portion **222** is formed with protruding from the inner surfaces of the side walls **23** in the width direction Y. When the position assurance member **3** is assembled, the guide portion **222** guides the position assurance member **3** along the connection direction X. Further, the guide portion **222** functions as a stopper at the time of assembly of the position assurance member **3**. For example, an end surface **222A** of the guide portion **222** that is on the opposite side of a fitting surface **2A** comes into contact with the position assurance member **3** at the time of assembly of the position assurance member **3**, and arranges the position assurance member **3** at a regular lock position. The fitting surface **2A** is an end surface of the housing **2** in the connection direction X, and is a surface facing the counterpart connector **10** in a state in which the housing **2** is fitted with the counterpart connector **10**.

[0020] The housing **2** is provided with a lock member **24**. The lock member **24** is a member that locks the housing **2** with respect to the counterpart connector **10** when the connector **1** and the counterpart connector **10** are connected, and is a member that engages with the position assurance member **3**. The lock member **24** is provided in the assembling space **22**, for example, and is installed along the connection direction X.

[0021] The lock member **24** includes a locked portion **241**, a press portion **242**, an arm portion **243**, and an installation portion **244**. The locked portion **241** is a portion that engages with a lock portion **121** of the counterpart connector **10** and a lock portion **331** of the position assurance member **3**. The press portion **242** is a portion that releases locking of the locked portion **241** and the lock portion **121** by being pressed, and releases a lock state of the counterpart connector **10** and the housing **2**. The locked portion **241**, the press portion **242** and the installation portion **244** each extend in the width direction Y intersecting with the connection direction X, and are respectively connected to two arm portions **243**, for example. The two arm portions **243** are arranged with being separated in the width direction Y, and each extend in the connection direction X. The locked portion **241**, the press portion **242**, and the installation portion **244** are installed between the two arm portions **243**, and are arranged in the order of the installation portion **244**, the locked portion **241**, and the press portion **242** from the fitting surface **2A** side.

[0022] The locked portion **241** is provided between facing surfaces **243A** of the two arm portions **243**, for example. The press portion **242** is provided with protruding upward toward the height direction Z as compared with the locked portion **241**, and connected to a top surface **243B** of the two arm portions **243**, for example. The installation portion **244** is connected to the ends of the two arm portions **243** on the fitting surface **2A** side. The installation portion **244** is formed to be longer than a distance between the two arm portions **243**, and ends on both sides are connected to the side walls **23**. A columnar portion **25** is connected to a central position in the width direction Y of the installation portion **244**. The columnar portion **25** is a member extending along the height direction Z, and is provided between the installation portion **244** and a floor portion **221** of the assembling space **22**. The columnar portion **25** supports the installation portion **244** from below. More specifically, the lock member **24** is supported at three points on the fitting surface **2A** side, and strength is improved. In other words, the lock member **24** is supported by the two side walls **23** and the columnar portion **25** on the fitting surface **2A** side through the installation portion **244**, and has a structure less likely to be broken even if external force is received. For example, if the installation portion **244** is supported by the two side walls **23** at two points, the installation portion **244** has a structure of being likely to bend in the height direction Z, and strength to external force becomes lower. In contrast to this, in the connector **1** according to the present embodiment, by the lock member **24** being supported at three points, improvement in strength is achieved, and strength to external force can be improved.

[0023] The ends of the two arm portions **243** on the opposite side of the fitting surface **2A** are respectively connected to the side walls **23**. That is, the lock member **24** is supported on both sides corresponding to the fitting surface **2A** side and the opposite side of the fitting surface **2A**. Thus,

the strength of the lock member **24** is improved as compared with a case of a cantilevered structure. [0024] A recess portion **244A** is formed in the installation portion **244**. The recess portion **244A** is formed by denting a surface of the installation portion **244** that faces the counterpart connector **10**, along the connection direction X. For example, the recess portion **244A** is formed by denting the entire installation portion **244** in the connection direction X. Here, the entire includes approximate entire. By the recess portion **244A** being formed, the installation portion **244** of the lock member **24** has a structure less likely to receive external force. For example, even in a case where the connector **1** drops onto a floor, the installation portion **244** is unlikely to come into direct contact with a floor surface, and the lock member **24** has a structure less susceptible to impact.

[0025] As illustrated in FIGS. **1** to **3**, the position assurance member **3** has a configuration in which a guide arm **32** and a lock arm **33** protrude from a base portion **31** toward the connection direction X. The base portion **31** is a plate-like portion erected upward, and provided toward a direction intersecting with the connection direction X. Two guide arms **32** are provided, and extend in the connection direction X from both ends in the width direction Y of the base portion **31**. In a case where the position assurance member **3** is assembled to the assembling space **22** of the housing **2**, the guide arm **32** is inserted along the guide portion **222**.

[0026] The lock arm **33** is provided between the two guide arms **32** with protruding in the connection direction X. The lock arm **33** is an engagement portion that engages with the housing **2**. In a case where the position assurance member **3** is assembled to the assembling space **22**, the lock arm **33** is inserted between the two arm portions **243**. In the lock arm **33**, the lock portion **331** is formed. The lock portion **331** is a portion that engages with the locked portion **241** of the lock member **24** of the housing **2**. In a case where the position assurance member **3** is inserted into the housing **2** up to a tentative lock position (standby position), the lock portion **331** is arranged at a position in front of the locked portion **241**. In a case where the connector **1** is appropriately connected to the counterpart connector **10**, the position assurance member **3** becomes insertable into the housing **2** up to a regular lock position, and it becomes possible to cause a user to recognize that the connector **1** is appropriately connected to the counterpart connector **10**. At this time, by the insertion of the position assurance member **3**, the lock portion **331** crawls under the locked portion **241** to become movable with respect to the locked portion **241** toward the fitting surface **2A** side of the housing **2**. On the other hand, in a case where the connector **1** is not appropriately connected to the counterpart connector **10**, the position assurance member **3** cannot be inserted into the housing **2** up to the regular lock position. In other words, the lock portion **331** cannot crawl under the locked portion **241** to move toward the fitting surface **2A** side of the locked portion **241**.

[0027] As illustrated in FIG. **1**, the counterpart connector **10** has a tubular shape, and has a configuration in which a facing surface **11** is opened and the connector **1** can be inserted into an internal space **12**. On a ceiling surface of the internal space **12**, a lock portion (not illustrated) is formed. The lock portion is a protruding portion that engages with the locked portion **241** of the lock member **24**. By the locked portion **241** being locked to the lock portion, the connector **1** is prevented from being detached from the counterpart connector **10**. In the internal space **12**, a plurality of terminals (not illustrated) is provided.

[0028] As illustrated in FIG. **4**, in the lock member **24** of the housing **2**, a protrusion portion **27** protruding downward from the arm portion **243** is formed. Further, a protrusion **28** protruding from the floor portion **221** of the assembling space **22** is formed. The protrusion portion **27** and the protrusion **28** are portions that engage with the position assurance member **3**. In a case where the housing **2** is resin-molded, the connection direction X corresponds to a mold opening direction. In this case, when the protrusion portion **27** and the protrusion **28** are formed, for mold release, a hole **26** needs to be formed on the fitting surface **2A** side of the assembling space **22**. It is considered to open the entire portion on the fitting surface **2A** side between the lock member **24** and the floor portion **221**, but there is a concern that the support strength of the lock member **24** becomes lower.

Thus, in the connector **1** according to the present embodiment, by supporting the lock member **24** by providing the columnar portion **25**, the support strength of the lock member **24** is improved. [0029] As illustrated in FIG. 5, the position assurance member **3** is assembled to the housing **2**, and arranged at the regular lock position. At this time, the position assurance member **3** is separated from the lock member **24**, and has contact with the end surface **222A** of the guide portion **222**. More specifically, the position assurance member **3** is inserted from the tentative lock position to the regular lock position in the assembling space **22**, and by a contact surface **311** of the base portion **31** coming into contact with the guide portion **222**, the position assurance member **3** is arranged at the regular lock position. At this time, the position assurance member **3** is separated from the lock member **24**, and a clearance gap **G** is formed between the position assurance member **3** and the lock member **24**.

[0030] Next, the mechanical strength of the connector **1** according to the present embodiment will be described.

[0031] As illustrated in FIG. 6, the connector **1** is connected to the counterpart connector **10** by assembling the position assurance member **3** to the housing **2**. The position assurance member **3** is inserted into the housing **2** up to the tentative lock position. An electric wire (not illustrated) is connected to the connector **1** and the counterpart connector **10**.

[0032] At this time, as illustrated in FIG. 7, the connector **1** sometimes bumps into an external object. For example, it is considered that the connector **1** erroneously drops and collides with a floor surface **80**. In a case where the connector **1** drops with the fitting surface **2A** being oriented downward, the fitting surface **2A** collides with the floor surface **80**. In this case, the connector **1** receives impact due to dropping, but large impact is prevented from being added to the lock member **24**. That is, the recess portion **244A** is formed in the installation portion **244** of the lock member **24**, and the lock member **24** is provided with being recessed from the fitting surface **2A**. Thus, in the connector **1** according to the present embodiment, the lock member **24** is prevented from coming into direct contact with the floor surface **80**, and impact to be received from the floor surface **80** can be reduced.

[0033] Further, the lock member **24** is supported at three points in the housing **2**, and mechanical strength is improved. That is, the installation portion **244** of the lock member **24** is connected with the side walls **23** on both sides, and connected with the columnar portion **25** extending from the floor portion **221** of the assembling space **22**. Thus, the installation portion **244** has a structure difficult to move in the width direction **Y** and the height direction **Z**, and a structure less likely to deform in a case where impact is received. Accordingly, in the connector **1** according to the present embodiment, even in a case where impact is received from the floor surface **80**, it is possible to prevent the lock member **24** from being easily broken.

[0034] Further, as illustrated in FIG. 5, in the connector **1**, in a case where the position assurance member **3** is assembled to the housing **2** and is arranged at the regular lock position, the position assurance member **3** has contact with the guide portion **222** of the assembling space **22**. At this time, the clearance gap **G** is formed between the position assurance member **3** and the lock member **24**, and the position assurance member **3** is separated from the lock member **24**. Thus, even in a case where the connector **1** comes into contact with an external object, direct impact is prevented from being added to the lock member **24**. In other words, even if the position assurance member **3** of the connector **1** comes into contact with the external object, impact is transmitted to the guide portion **222** of the housing **2**, and the impact is unlikely to be transmitted to the lock member **24**. In the connector according to the present embodiment, external force is prevented from being added from the lock member **24**, and bearing force for external force can be improved.

[0035] As described above, in the connector **1** according to the present embodiment, by the lock member **24** being supported at three points in the housing **2**, the mechanical strength of the housing **2** can be improved. That is, in the housing **2**, while the lock member **24** has a structure of being installed on the two side walls **23**, the lock member **24** is connected with the two side walls **23** and

the columnar portion 25, to be supported at three points. Thus, in the connector 1 according to the present embodiment, the lock member 24 can be strongly supported, and improvement in mechanical strength can be achieved.

[0036] Further, in the connector 1 according to the present embodiment, by forming the recess portion 244A in the installation portion 244 of the lock member 24, it is possible to prevent external force from being directly added to the lock member 24. With this configuration, in the connector 1 according to the present embodiment, it is possible to reduce impact to be added to the lock member 24.

[0037] Further, in the connector 1 according to the present embodiment, the position assurance member 3 is separated from the lock member 24 and brought into contact with the guide portion 222 at the regular lock position. Thus, in the connector 1 according to the present embodiment, in a case where the position assurance member 3 receives external force, it is possible to prevent external force from being transmitted to the lock member 24 via the position assurance member 3. Accordingly, the connector 1 according to the present embodiment can achieve improvement in mechanical strength.

[0038] In addition, the connector according to the present invention is not limited to the above-described embodiment, and various changes can be made within the scope set forth in the appended claims. The connector according to the present embodiment may be formed by appropriately combining the components of the above-described embodiment and a modified example.

[0039] For example, in the above-described embodiment, a case where the recess portion 244A is formed in the installation portion 244 has been described, but the recess portion 244A is not formed in the installation portion 244 in some cases. Even in this case, by supporting the lock member 24 at three points, it is possible to improve the strength of the connector 1.

[0040] According to the connector according to the present embodiment, it is possible to achieve improvement in strength.

[0041] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

## Claims

1. A connector comprising: a housing configured to be connectable to a counterpart connector, and form an assembling space along a connection direction; and a position assurance member to be inserted into the assembling space and assembled to the housing in such a manner as to be movable along the connection direction from a tentative lock position to a regular lock position, wherein the assembling space is formed between two side walls formed in the housing and extending in the connection direction, the housing includes a lock member that engages with the position assurance member assembled to the assembling space, and locks the position assurance member at the regular lock position, the lock member includes an installation portion installed between the two side walls, and the installation portion is connected to the two side walls, and connected to a columnar portion extending from a floor portion of the assembling space, to be supported at three points.
  2. The connector according to claim 1, wherein the installation portion is provided on a side of a fitting surface of the housing that faces the counterpart connector, and a recess portion is formed by denting a surface facing the counterpart connector, along the connection direction.
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